

Table 1 Determining minimum and maximum attainable values of correlation between two ordered categorical variables X and Y given their marginal distributions.

(a) Observed frequency distributions

	Category 1	Category 2	Category 3	Category 4	Total	
X	6	3	2	1	12	
Y	3	5	2	2	12	

(b) Comonotonic arrangement and pairing:

X: 1 1 1 1 1 1 2 2 2 3 3 4

Y: 1 1 1 2 2 2 2 2 3 3 4 4 $r_{\max} = 0.87831$

(c) Anti-comonotonic arrangement and pairing:

X: 1 1 1 1 1 1 2 2 2 3 3 4

Y: 4 4 3 3 2 2 2 2 2 1 1 1 $r_{\min} = -0.79466$

Table 2 Illustrative pairs of marginal distributions and their associated minimum and maximum attainable values of r ($r_{\min}$, $r_{\max}$) given these distributions. The distributions are given in counts are given ($n=12$). The categories are assigned consecutive integer values starting with 1. $C1$ and $C2$ are measures of constraint, as discussed in the text.

		Category 1	Category 2	Category 3	Category 4	Total	rmin	rmax
(a)	Two arbitrarily distributed ordinal categorical variables (orcats)							
	X	6	3	2	1	12	-0.79466	0.87831
	Y	3	5	2	2	12		
(b)	One arbitrarily distributed variable, and one with same distribution inversed							
	X	6	3	2	1	12	-1	0.71429
	Y	1	2	3	6	12		
(c)	Two variables with arbitrary identical distributions							
	X	6	3	2	1	12	-0.71429	1
	Y	6	3	2	1	12		
(d)	Two variables with identical symmetric and unimodal distributions							
	X	2	4	4	2	12	-1	1
	Y	2	4	4	2	12		
(e)	Two variables with different distributions, both symmetric and unimodal							
	X	2	4	4	2	12	-0.91169	0.91169
	Y	1	5	5	1	12		
(f)	Two variables with identical symmetric bimodal distributions							
	X	4	2	2	4	12	-1	1
	Y	4	2	2	4	12		
(g)	Two variables with different distributions, both symmetric and bimodal							
	X	4	2	2	4	12	-0.95673	0.95673
	Y	5	1	1	5	12		
(h)	One symmetric unimodal variable, the other symmetric bimodal							
	X	2	4	4	2	12	-0.89923	0.89923
	Y	4	2	2	4	12		
(i)	Two variables, one very left skewed, the other very right skewed							
	X	1	1	1	9	12	-0.95818	0.31939
	Y	8	2	1	1	12		

(j)	Same as situation (i), above, but categories assigned values 1; 2.25; 2.75; 4							
	X	1	1	1	9	12	-0.93321	0.33829
	Y	8	2	1	1	12		